

Timing graph

Static timing starts with a directed timing graph

Goldens to remember

- Goldens: six pins, five arcs, levels zero through five, path delay three point two
- A cycle returns no levels
- Keep these numbers handy, the browser challenges and Track A tests use the same instance

Pseudocode

- Pseudocode for this lab is Kahn levelize on a pin and arc timing graph
- Inputs are pins and delayed arcs
- The loop peels indegree-zero pins and writes levels
- Stop with failure if a cycle leaves pins unvisited
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On the tiny chain the sketch returns levels zero through five with out at five
- Six pins and five arcs are the shared instance
- Adding out to in makes levelize fail, that is the cycle golden

Algorithm sketch — try these

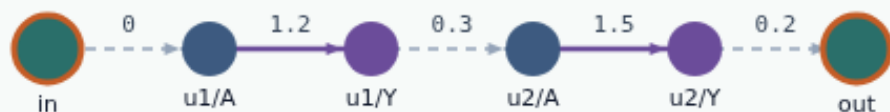
```
INPUT: pins, arcs (delay, kind cell|net)
OUTPUT: levels[] or FAIL(cycle)
indeg[v] ← |preds|; Q ← {v | indeg[v] = 0}; level[Q] = 0
while Q:
    u ← pop; for v in succ(u):
        indeg[v] -= 1; level[v] ← max(level[v], level[u] + 1)
        if indeg[v] = 0: push v
FAIL if not all visited else return levels
GOLDEN: 6 pins, 5 arcs; in:0 ... out:5
```

Start with a tiny timing chain

TIMING GRAPH

Start with a tiny timing chain

Step 1 / 5 · tiny-chain · module01-01-timing-graph



Explanation

Six pins form one path from primary input in through two cells to out. Net arcs are dashed; cell arcs are solid. This is the shared starter for every STA lab.

- Pins = ports and cell pins
- Directed arcs carry delay
- One path keeps goldens easy to check by hand

Pins: 6

Arcs: 5 (2 cell / 3 net)

Clock period: 10 (for later labs)

Start with a tiny timing chain

Name sources and sinks

TIMING GRAPH

Name sources and sinks

Step 2 / 5 · sources-sinks · module01-01-timing-graph



Explanation

A source has no incoming arc; a sink has no outgoing arc. On this chain, in is the only source and out is the only sink.

- Source = no predecessor arc
- Sink = no successor arc
- Arrival starts at sources; required at sinks

```
sources: in  
sinks: out
```

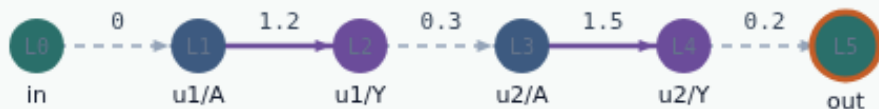
Name sources and sinks

Levelize with Kahn's algorithm

TIMING GRAPH

Levelize with Kahn's algorithm

Step 3 / 5 · levelize · module01-01-timing-graph



Explanation

Process pins with indegree zero, then peel successors.
out lands at level 5 on this chain.

- Queue pins with indegree 0
- $\text{level}(v) = 1 + \max(\text{level of preds})$
- If some pins never finish, there is a cycle

```
in: L0  
u1/A: L1 · u1/Y: L2  
u2/A: L3 · u2/Y: L4  
out: L5
```

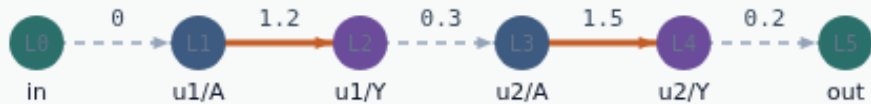
Levelize with Kahn's algorithm

Topo order and delay sums

TIMING GRAPH

Topo order and delay sums

Step 4 / 5 · topo-delays · module01-01-timing-graph



Explanation

Sorting pins by level gives the topo order. Path delay sums to 3.2; cell arcs alone sum to 2.7.

- Topo order follows increasing level
- Path delay Σ = sum of every arc
- Cell delay Σ ignores net arcs

```
topo: in → u1/A → u1/Y → u2/A → u2/Y → out  
path delay  $\Sigma$ : 3.2  
cell delay  $\Sigma$ : 2.7
```

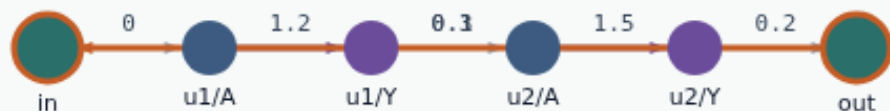
Topo order and delay sums

A cycle fails levelize

TIMING GRAPH

A cycle fails levelize

Step 5 / 5 · cycle-reject · module01-01-timing-graph



Explanation

Add a back-edge from out to in and Kahn's algorithm stalls. Timing graphs for STA must be acyclic.

- Back-edge out → in creates a cycle
- levelize returns null
- Fix the graph before propagating arrival

```
acyclic: false  
levels: null
```

A cycle fails levelize

Browser lab track

- In the browser lab, open **timing-graph**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run ``python3 common/test_propagate.py`` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue